

GSE183904 gastric cancer dataset4 - KCN intestinal vs diffuse/EMT pipeline

Biological question: are KCNQ1, KCNE2 and KCNE3 preferentially associated with malignant intestinal-like or diffuse/EMT-like gastric cancer cells?

1. Public dataset and biological scope

GSE183904, Kumar et al., Cancer Discovery 2022, DOI 10.1158/2159-8290.CD-21-0683. Data type: author-processed raw scRNA-seq gene-count matrices from Cell Ranger. Selected scope: primary tumors with Lauren label plus matched normals for CNA reference.

2A. Samples retained

20 primary tumors: 14 intestinal and 6 diffuse. 5 matched primary normal samples used only as inferCNV reference. Excluded: mixed tumors, metastases, peritoneal samples and missing Lauren labels.

2B. Input matrix

Compressed CSV count matrices inside GSE183904_RAW.tar. Gene x cell raw UMI counts streamed sample-by-sample and exported as sparse Matrix Market after epithelial candidate selection.

3. Broad epithelial candidate gate

Rule: ≥ 3 core epithelial genes, or ≥ 2 core epithelial genes with epithelial score $> 65\%$ of strongest non-epithelial score. Core genes include EPCAM, KRT7/8/18/19/20, MUC1, CDH1, TACSTD2. Output: 28,845 / 113,470 cells.

4. Seurat representation

LogNormalize scale factor 10,000; 2,000 variable genes; ScaleData regressing total UMI; PCA 30 components; neighbors/t-SNE on PCs 1-20; k=30; resolution=0.8; seed=12345.

5. RNA-inferred CNA support

inferCNV using 750 matched-normal epithelial reference cells, balanced as 150 cells x 5 normal patients. GRCh38 gene order; cutoff=0.1; denoise=TRUE; HMM=FALSE; 8 threads; high CNA threshold = normal 95th percentile, 0.0552533.

6. Malignant epithelial selection

Tumor and non-malignant epithelial programs from Zhou et al. Supp. Table S4. Primary rule: tumor sample and tumor-program difference $>$ normal 95th percentile, 1.10733. 8,543 selected by tumor program + 2,785 rescued by inferCNV. After removing 2,427 immune-contaminated cells: 8,901 malignant epithelial candidates.

7. Intestinal vs diffuse/EMT scoring

UCell max rank=1,500. Intestinal signature: 22 genes. Diffuse/EMT score = $z(36 \text{ EMT-up genes}) - z(10 \text{ epithelial-junction genes})$. KCNQ1, KCNE2 and KCNE3 are absent from classifiers. High-confidence rule: $\text{abs}(\text{delta}) \geq 0.5 \text{ SD}$ and dominant score ≥ 0 .

8. Final high-confidence cells

6,573 final malignant epithelial cells: 4,170 intestinal-like, 2,403 diffuse/EMT-like, 2,328 indeterminate excluded. Lauren sanity check: diffuse patients have higher median diffuse/EMT-like fraction, 0.670 vs 0.289; Wilcoxon $p=0.0433$.

9. KCN tests

Detection = raw count ≥ 1 . Two-sided Fisher exact test compares expressing vs non-expressing cells; BH FDR across 3 KCN genes. Continuous association: Spearman on normalized expression, plus within-patient rank permutation with 1,000 permutations.

Cell-flow metrics retained in the final analysis

Selected GEO matrices	113,470 cells	primary tumors + matched normals
Broad epithelial candidates	28,845 cells	relaxed epithelial gate
Malignant epithelial candidates	11,328 cells	tumor program + inferCNV rescue
After immune-contamination filtering	8,901 cells	immune marker threshold ≥ 2 removed
Final high-confidence states	6,573 cells	4,170 intestinal-like; 2,403 diffuse/EMT-like

Final KCN readout in high-confidence malignant states

Gene	Detection Int.	Detection Diff.	Raw counts Int./Diff.	Fisher FDR	Direction
KCNQ1	25.7%	1.7%	1714 / 46	7.1e-178	intestinal
KCNE2	4.1%	0.8%	316 / 54	3.5e-17	weak intestinal
KCNE3	28.2%	5.7%	2282 / 190	1.4e-122	intestinal

Patient-aware Spearman supports KCNQ1 and KCNE3 intestinal orientation; KCNE2 remains low and less robust.